



Disable Touch ID in iOS 11

A new feature with iOS 11 allows users to easily change settings so your fingers can't unlock your phone using Touch ID. <http://dgit.in/Touchi0S>



Ban on killer robots

AI leaders from around the world including Elon Musk and Google signed a new agreement banning the use of robots in military. <http://dgit.in/KillerBot1>

Mucking about with DNA

Sequencing genomes is all well and good, but if you had ever wondered what that information enables us to do, here are a few crazy things that are now possible

Prithvi Sudan | feedback@dgit.in

H

umanity was forever set on an irreversible path when a curious Austrian started messing around with

his garden plants. Mendel and his peas may have been a very basic and rudimentary grasp at understanding genetics but today, with all the modern tools and technology available to geneticists, they are able to experiment with accuracy and precision that is of several orders of higher magnitude than ever before. We are now able to actively change the DNA of organisms to achieve lots of interesting and downright baffling things. One of humanity's greatest yet relatively unsung achievements is the development of CRISPR-Cas9. In fact, many of these achievements below wouldn't be possible without it. If you want to learn more about this technology, here's some background reading: <http://dgit.in/>

CrisprDNA. So here are some of the coolest things geneticists have achieved quite recently.

SEMI-SYNTHETIC ORGANISMS

Quick biology question: How many nucleobases is DNA made of? The correct answer is four – A, C, G, T. But now, scientists have successfully created bacteria that has DNA with 6 bases (the extra two being X and Y). “Why is that interesting?” you may ask, and it's interesting, very interesting in fact, because it gives us a glimpse of how life actually formed out of the building blocks that were present in Earth's soup of creation. It may even lead to humans being able to create synthetic life! Oh, the humanity!

Scripps Research Institute in California has had a team working on this exact same problem for a couple of years now (<http://dgit.in/ItsAlivee>). They had their first breakthrough in 2014, where they discovered the X and Y nucleobases. At that time, inserting them into an organism and hoping

for it to survive was just a theoretical possibility but today, they have managed to achieve it. The team, headed by Dr. Floyd Romesberg, have been experimenting with a particular type of bacteria called E. Coli (it's commonly found in your gut and is generally harmless; beneficial even, because it synthesises the vitamin B2, but some strains can cause you to have severe food poisoning).

They genetically modified the E. Coli to accept the X and Y bases as a part of their natural makeup and hold onto them for their whole lifetime. The real challenge was to keep the E. Coli from rejecting the new bases after some growth – all the early experiments ended up in failure because X and Y got rejected or the resulting organism was too weak and sickly to survive. Thanks to CRISPR, they reprogrammed the bacteria to not recognise the new bases as foreign bodies and this allowed the bacteria to better assimilate the bases. Right now, the bases don't really do anything but that could change in the future – scientists can program different attributes into those bases and that could result in a totally different organism.

And that does sound slightly spooky. What if some mad scientist can now genuinely cackle in glee and cry